A Transaction: All or Nothing

Safely combining database changes

SCAN / LEARN

THE MAIN IDEA

A database transaction groups related changes into one unit. If a required step fails, the system can undo the unfinished unit instead of leaving half a result.

FOLLOW THE SYSTEM

01 BEGIN

A customer transfers money.

02 DEBIT

The sender account is reduced.

03 CREDIT

The receiver account is increased.

04 COMMIT / ROLLBACK

Save both changes, or undo the transaction.

BEGIN → UPDATE A → UPDATE B → COMMIT

COMMON MISCONCEPTION

Writing two lines of database code makes the changes inseparable.

MORE ACCURATE

A transaction and correct isolation settings are needed.

THREE QUESTIONS TO KEEP

PROBLEM: How do related updates succeed together?

FAILURE: Half-completed operations corrupt records.

IN REAL LIFE: Bank transfers, orders and reservations.

SOLO DISCOVERY

Invent a two-step booking: what if the second step fails?